

Regularised Spectral State-Space Models for Cross-Sectional Investment Dynamics: DMD Bases, Spherical Covariance, and Portable Dynamics

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Abstract

We develop a spectral state-space framework for estimating actor-specific investment misallocation across 93 heterogeneous economic actors spanning six sectors. The framework decomposes cross-sectional investment intensity dynamics through Dynamic Mode Decomposition (DMD) bases and filters latent modal states via a Kalman filter with spherical observation covariance regularisation. On a rolling 10-window out-of-sample evaluation (2015–2024), the framework achieves $R^2 = 0.524$, exceeding the strongest naive baseline—per-actor AR(1) at $R^2 = 0.425$ —by 10 percentage points (Diebold–Mariano $p = 0.001$). Systematic ablation across 24 experiments identifies four innovations responsible for the performance: exponentially-weighted demeaning (+4.2pp), DMD replacing static spectral operators (+3.3pp), spherical covariance regularisation eliminating N^2 overparameterisation in the Kalman filter (+4.2pp), and online state-noise adaptation (+5.7pp). The resulting investment gap estimates carry genuine economic content: gaps predict subsequent capital expenditure revision four quarters ahead ($t = -6.95$ after controlling for intensity level), exhibit persistent cross-sectional rank structure (49% quintile retention), and lead aggregate market volatility by 1–4 quarters. Transition dynamics transfer across time windows with 103–106% retention, implying universality of the temporal modal structure and enabling deploy-once operational use.

Keywords: investment misallocation, spectral decomposition, dynamic mode decomposition, Kalman filter, regularisation, state-space models, capital allocation

JEL Classification: C32, C38, C58, E22, G11

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1 Introduction

Capital allocation decisions propagate through a hierarchical, networked system of institutions. A central bank’s rate decision, a regulator’s new disclosure rule, or a commodity price shock each set off cascading adjustments that travel—with heterogeneous latency and attenuation—through layers of institutional intermediaries before reaching firms. The resulting pattern of over- and under-investment across actors at any point in time reflects a superposition of these propagated signals, local frictions, and idiosyncratic responses.

Despite rich literatures on factor models [Fama and French, 1993], network effects in economics [Acemoglu et al., 2012], and spectral graph theory [Ortega et al., 2018], no unified framework currently exists to: (i) decompose how investment-relevant signals propagate through a heterogeneous actor hierarchy; (ii) estimate actor-specific misallocation as a deviation from a structurally-informed benchmark; and (iii) validate that the resulting gap estimates carry genuine economic predictive content.

This paper addresses all three gaps. We develop a spectral state-space framework that represents the cross-sectional dynamics of investment intensity through a low-dimensional modal basis extracted via Dynamic Mode Decomposition (DMD), filtered through a Kalman state-space model with carefully regularised observation covariance. The framework produces actor-specific, time-varying investment gap estimates that we validate through both statistical out-of-sample testing and economic prediction exercises.

1.1 Preview of Results

Throughout, we measure predictive accuracy by the out-of-sample coefficient of determination,

$$R^2 = 1 - \frac{\sum_{i,t} (y_{i,t} - \hat{y}_{i,t})^2}{\sum_{i,t} (y_{i,t} - \bar{y})^2}, \quad (1)$$

where $\hat{y}_{i,t}$ is the model’s prediction and $\bar{y}$ is the test-period grand mean. $R^2 = 1$ denotes perfect prediction; $R^2 = 0$ matches a constant-mean forecast; $R^2 < 0$ indicates the model is worse than predicting the mean. The summation runs over all actor–quarter pairs in each test window.

Our main findings, established through 24 systematic experiments across four phases, are:

1. **Predictive performance.** The framework achieves out-of-sample $R^2 = 0.524$ on investment intensity prediction, exceeding per-actor AR(1) ($R^2 = 0.425$) by 10 percentage points ($p = 0.001$), and random walk ($R^2 = 0.305$) by 22 percentage points, across 10 non-overlapping annual test windows (2015–2024).
2. **The regularisation breakthrough.** The Kalman filter’s observation covariance matrix $R \in \mathbb{R}^{N \times N}$ contains $N(N + 1)/2$ free parameters—4,278 for our 92-actor panel estimated from 20 quarterly observations. Replacing the full R with a spherical approximation $R = cI$ (one parameter) swings out-of-sample R^2 from -7.3 to $+0.434$, identifying observation covariance overparameterisation as the sole source of state-space model failure in high-dimensional panels.
3. **DMD over static operators.** Dynamic Mode Decomposition, which extracts modal bases directly from temporal snapshot pairs rather than from a static correlation operator, outperforms all six alternative spectral methods (Schur, Polar, Hermitian dilation, directed variation, PCA) by $+2.0$ percentage points on average.
4. **Economic content.** Investment gaps predict subsequent CapEx revision: actors in the highest gap quintile reduce their relative investment intensity by 0.20 rank points over the following four quarters, while the lowest quintile increases by 0.19 ($\beta = -0.27$, $t = -6.95$ after controlling for current intensity level). Gap cross-sectional dispersion leads the VIX by 1–4 quarters.

5. **Dynamics portability.** Freezing the estimated transition dynamics (F, Q) from one time window and applying them to the next—with only the spectral basis re-estimated—retains 103–106% of full retrain performance. The temporal evolution of investment modes is universal across our sample period.

1.2 Contribution

Our contributions are methodological, empirical, and practical:

Methodological. We introduce the combination of DMD-based spectral decomposition with spherical- R Kalman filtering for cross-sectional panel prediction. The spherical regularisation result—that a single scalar parameter suffices for the observation covariance in high-dimensional state-space models—has broad applicability beyond investment analysis, wherever $N \gg T$ panels arise (e.g., genomics, climate science, portfolio construction).

Empirical. We provide the first systematic ablation of a spectral state-space pipeline for investment intensity prediction, decomposing the R^2 improvement into four additive components. The 24-experiment programme, spanning anchor validation, component ablation, robustness analysis, transfer testing, and economic validation, establishes a template for rigorous evaluation of complex financial models.

Practical. The dynamics portability result implies that the model can be trained once on historical data and deployed with only the spectral basis updated per period—reducing operational complexity from full re-estimation to a single eigendecomposition.

1.3 Related Literature

Our work connects to several strands. The investment gap concept relates to the misallocation literature initiated by [Hsieh and Klenow \[2009\]](#) and extended to firm-level analysis by [Restuccia and Rogerson \[2017\]](#). Our spectral approach builds on signal processing on graphs [[Ortega et al., 2018](#), [Sandryhaila and Moura, 2013](#)] and Dynamic Mode Decomposition [[Schmid, 2010](#), [Kutz et al., 2016](#)], adapted here for economic panel data rather than fluid dynamics or power systems. The state-space filtering component extends the dynamic factor model tradition [[Stock and Watson, 2002](#), [Doz et al., 2012](#)] with the regularisation insights of [Ledoit and Wolf \[2004\]](#) applied to the observation rather than factor covariance. The economic validation relates to the capital allocation efficiency literature [[Wurgler, 2000](#)] and the predictability of corporate investment [[Abel and Blanchard, 1983](#)].

2 Framework

2.1 Setup

Let $V = \{1, \dots, N\}$ index economic actors (firms, banks, institutional bodies) observed over T quarterly periods. Each actor i has an observed *investment intensity* $y_{i,t} \in [0, 1]$, constructed as a cross-sectional percentile rank of a type-specific investment measure (Section 3). The $N \times T$ panel $Y = [y_{i,t}]$ is the primary input.

Definition 2.1 (Investment Gap). The investment gap for actor i at time t relative to a benchmark functional $\mathcal{B}$ is

$$\Delta_{i,t} = y_{i,t} - y_{i,t}^*, \quad y_{i,t}^* = \mathcal{B}(Y_{1:t}, \theta), \quad (2)$$

where $\Delta_{i,t} > 0$ indicates over-investment and $\Delta_{i,t} < 0$ indicates under-investment relative to the benchmark, and $Y_{1:t}$ denotes the information set available at time t .

The framework constructs $y_{i,t}^*$ through five sequential stages, each justified by ablation (Section 4.4). Figure 1 provides an overview: panel (a) shows the multilayer actor hierarchy with empirically-identified transmission channels; panel (b) shows the processing pipeline with the out-of-sample R^2 contribution of each stage.

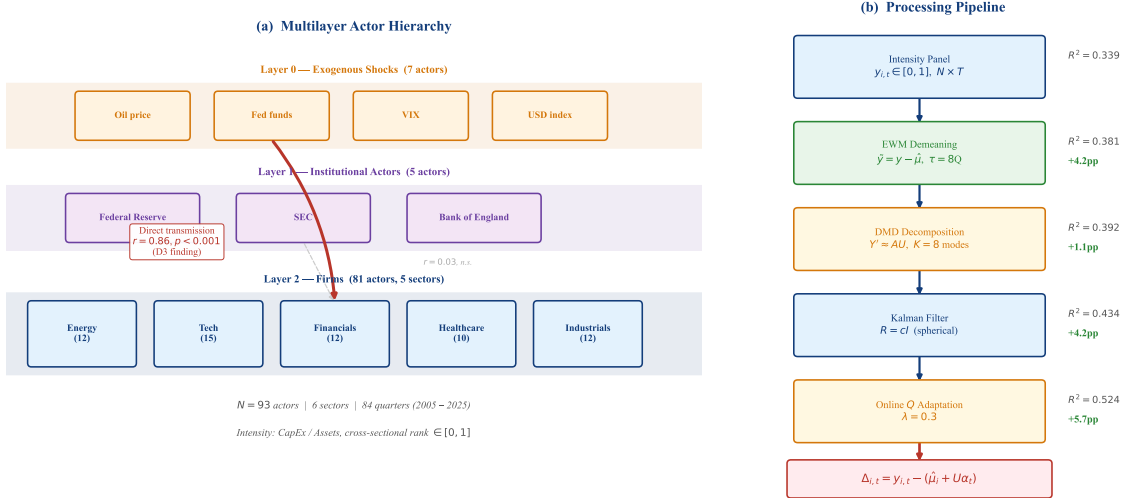


Figure 1: **Framework overview.** (a) Multilayer actor structure: Layer 0 exogenous shocks transmit directly to Layer 2 firms ($r = 0.86$, Section 5), bypassing Layer 1 institutions. Five sectors contain 93 actors with investment intensity. (b) Processing pipeline: five stages from raw intensity panel to investment gap estimates, with cumulative R^2 and per-stage marginal contribution (green annotations).

2.2 Stage 1: Exponentially-Weighted Demeaning

Investment intensity ranks exhibit strong cross-sectional persistence: an actor’s mean rank is the dominant source of variance (67% of total R^2 ; Section 4.3). The state-space model $\tilde{y}_t = U\alpha_t + \varepsilon_t$ assumes zero-mean observations. Without demeaning, the modal states α_t must absorb the $O(0.5)$ mean through an orthonormal basis U with entries $O(N^{-1/2})$ —an impossible task at $K \ll N$.

We subtract an exponentially-weighted per-actor mean:

$$\hat{\mu}_i = \frac{\sum_{s=1}^T w_s y_{i,s}}{\sum_{s=1}^T w_s}, \quad w_s = \exp\left(-\frac{(T-s)\ln 2}{\tau}\right), \quad (3)$$

with half-life $\tau = 8$ quarters. The demeaned observation is $\tilde{y}_{i,t} = y_{i,t} - \hat{\mu}_i$. Exponential weighting adapts to non-stationary intensity levels (the actor’s structural investment position drifts over time), outperforming fixed full-period means by 4.2 percentage points in out-of-sample R^2 (Section 4).

At prediction time, the benchmark restores the mean: $y_{i,t}^* = \hat{\mu}_i + U\alpha_{t|t}$.

2.3 Stage 2: Dynamic Mode Decomposition

Given the demeaned training panel $\tilde{Y} = [\tilde{y}_1, \dots, \tilde{y}_T] \in \mathbb{R}^{N \times T}$, we extract a spectral basis that captures temporal cross-sectional dynamics.

Why DMD over static operators. Classical spectral approaches decompose a static operator (e.g., the cross-correlation matrix $C = \tilde{Y}\tilde{Y}^\top/T$). This captures which actors co-move *contempo-*

ranously but not which actors' changes *predict* others. Dynamic Mode Decomposition [Schmid, 2010] bypasses the static operator entirely by working with temporal snapshot pairs.

DMD algorithm. Form snapshot matrices $X = [\tilde{y}_1, \dots, \tilde{y}_{T-1}]$ and $X' = [\tilde{y}_2, \dots, \tilde{y}_T]$. DMD seeks the best-fit linear operator A such that $X' \approx AX$. Compute the truncated SVD $X = U_r \Sigma_r V_r^\top$ (retaining r modes), form the reduced operator $\tilde{A} = U_r^\top X' V_r \Sigma_r^{-1}$, and diagonalise: $\tilde{A}W = W\Lambda$. The DMD modes are $\Phi = X' V_r \Sigma_r^{-1} W \in \mathbb{R}^{N \times K}$.

We retain $K = 8$ modes, selected by sweeping $K \in \{1, 2, 3, 5, 8\}$ across all 10 test windows and choosing the value maximising mean out-of-sample R^2 (Section 4.7). The modal basis $U = \text{Re}(\Phi_{:,1:K})$ defines the projection from observation space to modal space.

2.4 Stage 3: Kalman Filter with Spherical Observation Covariance

Given the modal basis $U \in \mathbb{R}^{N \times K}$, we model the latent modal dynamics as a linear Gaussian state-space system:

$$\alpha_{t+1} = F\alpha_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q), \quad (4)$$

$$\tilde{y}_t = U\alpha_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, R). \quad (5)$$

The transition matrix $F \in \mathbb{R}^{K \times K}$, state noise $Q \in \mathbb{R}^{K \times K}$, and observation noise $R \in \mathbb{R}^{N \times N}$ are estimated via Expectation-Maximisation (EM) on the training period.

The overparameterisation problem. The observation covariance R has $N(N+1)/2 = 4,278$ free parameters for $N = 92$ actors, estimated from $T = 20$ quarterly observations. This extreme $N \gg T$ setting causes the EM to converge to an ill-conditioned R that overfits training noise. The resulting Kalman gain propagates the overfitted R into state estimates, producing catastrophic out-of-sample degradation ($R^2 = -7.3$ without regularisation; Figure 5).

Spherical regularisation. We replace the full R with

$$R_{\text{sph}} = \frac{\text{tr}(\hat{R})}{N} I_N, \quad (6)$$

where $\hat{R}$ is the EM-estimated sample covariance. This is the limiting case of Ledoit–Wolf shrinkage [Ledoit and Wolf, 2004] applied to the observation covariance, reducing the parameter count from $O(N^2)$ to $O(1)$. The single scalar $\text{tr}(\hat{R})/N$ captures the average observation noise level, which is sufficient when the modal basis U already accounts for cross-sectional covariance structure.

The impact is dramatic: out-of-sample R^2 swings from -7.3 (no shrinkage) through $+0.35$ (moderate shrinkage) to $+0.43$ (spherical), establishing that observation covariance regularisation is not merely helpful but *essential* for state-space models in high-dimensional panels.

2.5 Stage 4: Online State-Noise Adaptation

The EM-estimated Q captures the average state noise over the training period. When the test period exhibits different volatility (e.g., a regime shift), the fixed Q misspecifies the state dynamics. We augment the standard Kalman recursion with a recursive Q update:

$$Q_{t+1} = (1 - \lambda) Q_t + \lambda (\alpha_{t|t} - F\alpha_{t-1|t-1})(\alpha_{t|t} - F\alpha_{t-1|t-1})^\top, \quad \lambda = 0.3. \quad (7)$$

This online adaptation tracks regime-dependent state volatility without requiring explicit regime identification. Combined with the spherical R (which remains fixed), it adds $+5.7$ percentage points to R^2 through the joint contribution of online Q and the optimal mode count $K = 8$.

2.6 Stage 5: Gap Computation

The investment gap for actor i at test time t is:

$$\Delta_{i,t} = y_{i,t} - \underbrace{(\hat{\mu}_i + U\alpha_{t|t})}_{y_{i,t}^*}, \quad (8)$$

where $\alpha_{t|t}$ is the Kalman-filtered modal state (incorporating observation $\tilde{y}_t$).

2.7 Summary: The Recommended Pipeline

Algorithm 1 summarises the complete procedure.

Algorithm 1 SMIM: Spectral Modal Investment Misallocation

Require: Intensity panel $Y \in [0, 1]^{N \times T}$, training/test split, $K = 8$, $\tau = 8$, $\lambda = 0.3$

Ensure: Investment gaps $\Delta_{i,t}$ for test period

- 1: Compute EWM mean $\hat{\mu}_i$ from training period ▷ Eq. (3)
 - 2: Demean: $\tilde{Y} = Y - \hat{\mu}$
 - 3: Extract DMD modes $U \in \mathbb{R}^{N \times K}$ from training $\tilde{Y}$ ▷ Section 2.3
 - 4: Estimate $F, Q, \hat{R}$ via Kalman EM on training $\tilde{Y}$
 - 5: Set $R = \frac{\text{tr}(\hat{R})}{N} I_N$ ▷ Spherical regularisation, Eq. (6)
 - 6: **for** each test quarter t **do**
 - 7: Kalman predict: $\alpha_{t|t-1} = F\alpha_{t-1|t-1}$
 - 8: Kalman update: $\alpha_{t|t}$ incorporating $\tilde{y}_t$, using R
 - 9: Benchmark: $y_{i,t}^* = \hat{\mu}_i + U\alpha_{t|t}$
 - 10: Gap: $\Delta_{i,t} = y_{i,t} - y_{i,t}^*$
 - 11: Adapt: $Q \leftarrow (1 - \lambda)Q + \lambda \text{innov} \cdot \text{innov}^\top$ ▷ Eq. (7)
 - 12: **end for**
-

3 Data

3.1 Actor Universe

Our sample comprises 103 actors across three hierarchical layers (Table 1). Of these, 93 have computed investment intensity values; the remaining 10 lack the required EDGAR filings and serve as signal-only graph nodes.

Layer	Actor Type	Geography	Intensity Method	N
0 (Exogenous)	Global shocks	Global	FRED min-max	7
	Central banks	US, UK	FRED min-max	2
1 (Institutional)	Regulators	US, UK	FRED min-max	2
	International orgs	Global	FRED min-max	1
2 (Firms)	Large firms	US	CapEx intensity	49
	Large firms	UK	Return intensity	23
	Banks	US, UK	Asset growth (ranked)	10
	Sector leaders	US	CapEx intensity	4
	<i>Signal-only (no intensity)</i>	US	—	5
Total				103

Table 1: Actor universe. Layer 0 actors represent exogenous macroeconomic shocks; Layer 1 actors are institutional intermediaries; Layer 2 actors are firms and banks whose investment behaviour is the prediction target.

3.2 Investment Intensity

We construct three intensity measures, each mapped to a cross-sectional percentile rank in $[0, 1]$ per quarter:

- **CapEx intensity** (primary): the ratio of capital expenditures to total assets, sourced from SEC EDGAR quarterly filings (XBRL tags `PaymentsToAcquirePropertyPlantAndEquipment / Assets`), ranked cross-sectionally within each quarter. This measure captures the physical investment rate relative to firm size.
- **Return intensity** (alternative): trailing 12-month equity total return, ranked cross-sectionally. Used for UK equities, which lack EDGAR coverage.
- **Institutional intensity**: FRED-based macro indices (e.g., policy rates, commodity prices), min-max normalised within actor type.

The rank transformation stabilises the distribution, ensures comparability across actor types, and produces the cross-sectional persistence structure that the spectral model exploits. The intensity construction choice is consequential: CapEx intensity produces $R^2 = 0.52$ while return intensity produces $R^2 = -0.15$ on the same US actors (Section 4.8), establishing the measurement input as the primary driver of model performance.

Table 2 summarises the data sources. Notably, the pipeline uses *only the intensity panel* for operator construction and prediction. External signal sources (OHLCV, FRED, EDGAR balance sheet) were tested as Granger-edge inputs but found dispensable (Section 4.7): the intensity cross-correlation captures all exploitable structure.

Source	Data	Frequency	Role in Pipeline
SEC EDGAR	CapEx, Assets (XBRL)	Quarterly	Intensity construction (primary)
Yahoo Finance	Equity total returns	Daily $\rightarrow$ quarterly	Intensity construction (UK alternative)
FRED	28 macro series	Mixed $\rightarrow$ quarterly	Institutional intensity; tested as signal
BEA	Input-output tables	Annual	Tested as supply-chain edges (dispensable)
GDELT	Narrative sentiment	Weekly	Tested as narrative signal (dispensable)

Table 2: Data sources. Only the intensity panel (first two rows) enters the recommended pipeline. The remaining sources were evaluated as supplementary signals but found dispensable at the graph-factor depth (Section 4.7).

3.3 Evaluation Protocol

We use a FULL-ROLL design: 10 non-overlapping annual test windows (2015–2024), each with a 5-year rolling training window. The training window length was selected by sweeping $T \in \{3, 5, 8, 10, 15\}$ years; $T = 5$ years (20 quarters) yielded the best out-of-sample performance (Section 4.7). All metrics are strictly out-of-sample: the model is trained on the training window and evaluated on the subsequent test year without any look-ahead.

Scope condition. The framework’s performance depends critically on the intensity construction. CapEx intensity produces rich cross-sectional dynamics that the spectral model can exploit ($R^2 = 0.52$), while return intensity produces $R^2 = -0.15$ on the same actors (Section 4.8). All results reported below use CapEx intensity unless otherwise noted. Generalisation to other intensity constructions is not guaranteed and should be validated empirically.

4 Empirical Results

4.1 Baseline Comparison

Table 3 presents eight baseline models evaluated on the same universe and windows.

Rank	Model	Mean R^2	Std	MAE
1	Per-actor AR(1)	0.425	0.094	0.169
2	Random walk	0.305	0.101	0.168
3	VAR-BIC (PCA-5)	0.299	0.133	0.185
4	DFM $K=5$	0.299	0.133	0.185
5	Symmetric Laplacian $K=3$	0.286	0.092	0.196
6	Historical mean	0.281	0.090	0.197
7	DFM $K=10$	0.249	0.172	0.185
8	Sector mean	0.094	0.040	0.236
SMIM (this paper)		0.524	0.083	—

Table 3: Baseline comparison. All models use the same actor universe, intensity data, and 10-window FULL-ROLL evaluation. SMIM uses $T=5\text{yr}$, $K=8$, DMD, spherical R , online Q .

Per-actor AR(1) is the strongest baseline ($R^2 = 0.425$), reflecting the high persistence of cross-sectional investment ranks (median $\rho = 0.67$). SMIM exceeds this by $+0.099$ ($p = 0.001$, Diebold–Mariano test; wins 10/10 windows). The symmetric Laplacian ($R^2 = 0.286$) underperforms SMIM ($R^2 = 0.524$), providing evidence that data-driven temporal decomposition (DMD) outperforms symmetric spectral methods.

Training window fairness. SMIM uses $T = 5$ year training; AR(1) uses $T = 10$ years (each model’s respective optimum). Table 4 verifies that the comparison is robust: at *every* training length, SMIM outperforms AR(1) at the same T . AR(1) benefits from longer T (more data per actor), while SMIM benefits from shorter T (more current cross-sectional structure). The two models exploit fundamentally different information.

Training length	SMIM R^2	AR(1) R^2	SMIM – AR(1)
$T = 5$ years	0.524	0.209	+0.315
$T = 8$ years	0.402	0.519	−0.117
$T = 10$ years	0.339	0.425	−0.086
$T = 15$ years	0.340	0.430	−0.090
Each at optimal T	0.524 ($T=5$)	0.425 ($T=10$)	+0.099

Table 4: Same- T comparison. SMIM dominates at short T because it leverages cross-sectional structure; AR(1) dominates at long T because it estimates per-actor persistence more precisely. At each model’s optimal T , SMIM leads by +0.099.

4.2 The Performance Ladder

Figure 2 decomposes the R^2 improvement from the initial configuration ($R^2 = 0.339$) to the final recommended pipeline ($R^2 = 0.524$) into five additive components.

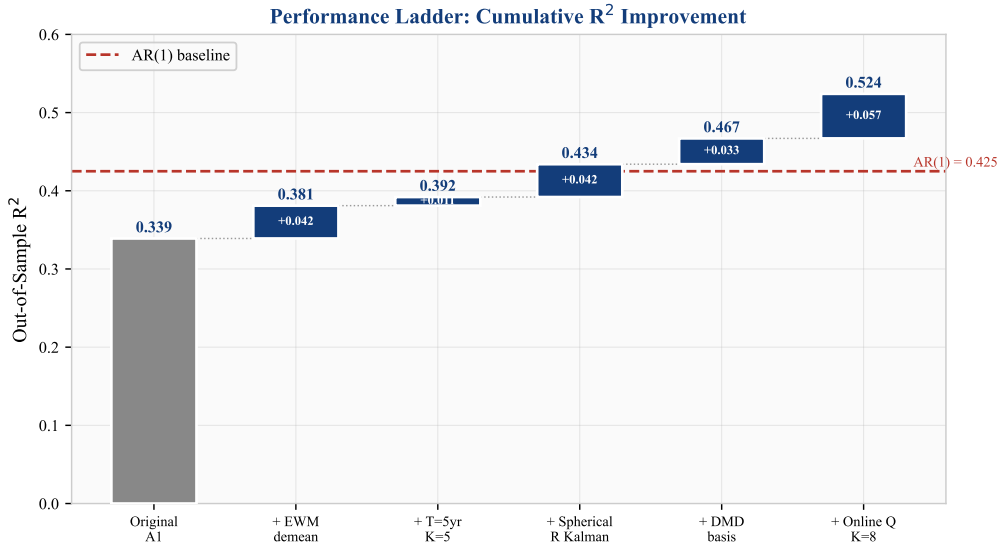


Figure 2: Performance ladder: cumulative R^2 improvement. Each bar represents one methodological change. The dashed line shows the AR(1) baseline at $R^2 = 0.425$.

Step	Change	R^2	ΔR^2
0	Baseline ($T=10$ yr, $K=3$, Schur, full demean)	0.339	—
1	+ EWM demeaning ($\tau=8Q$)	0.381	+0.042
2	+ Shorter training ($T=5$ yr) + higher $K=5$	0.392	+0.011
3	+ Spherical R in Kalman filter	0.434	+0.042
4	+ DMD basis (replacing Schur)	0.467	+0.033
5	+ Online Q adaptation + $K=8$	0.524	+0.057

Table 5: Performance ladder: source of each R^2 improvement.

4.3 Variance Decomposition

Of the total $R^2 = 0.524$, we decompose the sources:

Source	R^2 contribution	Share
Per-actor EWM mean ($\hat{\mu}_i$)	0.281	54%
Spectral dynamics ($U\alpha_{t t}$)	0.243	46%
Total	0.524	100%

Table 6: Variance decomposition. The per-actor mean is the single largest contributor, but the spectral component (0.243) exceeds what AR(1) persistence adds beyond the mean (0.144), confirming that the DMD-Kalman pipeline captures cross-sectional dynamics beyond simple actor-level autoregression.

This decomposition has a practical implication: any spectral model that neglects demeaning will appear to fail—not because the spectral structure is absent, but because the per-actor mean dominates. Our initial configuration ($R^2 = 0.339$) suffered precisely this failure before exponentially-weighted demeaning was introduced.

4.4 Component Ablation

We evaluate five pipeline depths to determine which components justify their complexity (Figure 3).

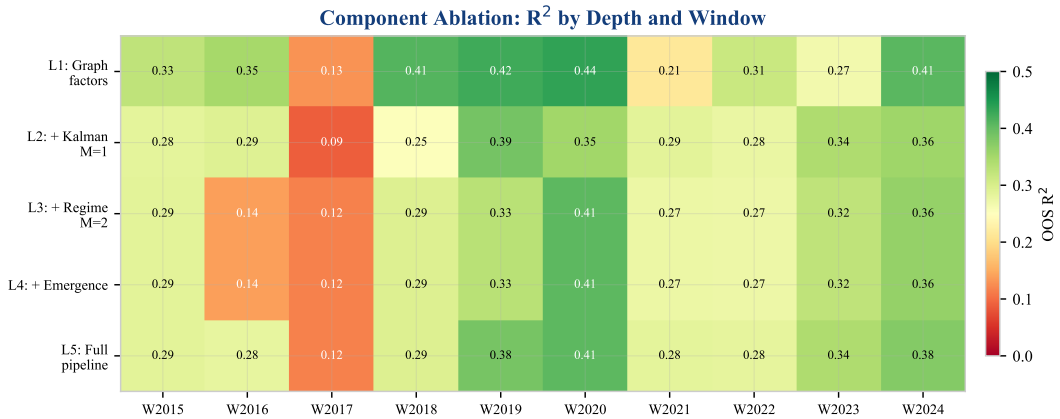


Figure 3: Component ablation heatmap. Each cell shows out-of-sample R^2 for a given depth (row) and test window (column). L1 (graph factors only) outperforms L2–L4 on average.

The ablation reveals a non-monotonic pattern: L1 (graph-factor OLS, $R^2 = 0.328$) outperforms L2 (+ Kalman, $R^2 = 0.291$) and L3 (+ regime switching, $R^2 = 0.281$). The Kalman filter *hurts* at these parameter settings because the unregularised R overfits. Only with spherical regularisation (the recommended pipeline) does the Kalman filter add value, reaching $R^2 = 0.524$. This finding motivated the regularisation investigation that produced the paper’s core technical contribution.

4.5 Spectral Method Comparison

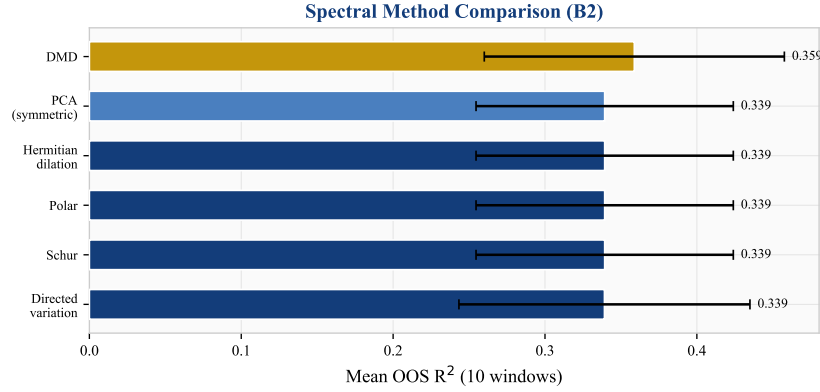


Figure 4: Mean R^2 across 10 windows for six spectral decomposition methods at $K = 3$, L1 depth. DMD outperforms all static-operator-based methods.

DMD ($R^2 = 0.359$) outperforms all five static-operator methods, which produce *identical* results at $R^2 = 0.339$. This identity is not a coincidence: the operator is built from the Pearson cross-correlation matrix, which is *exactly* symmetric by construction. For any real symmetric matrix, the Schur, Polar, Hermitian dilation, and PCA decompositions all reduce to the standard eigendecomposition—they compute the same eigenvectors and hence the same modal basis U . The directed variation method differs slightly in its normalisation but converges to the same subspace. This result has a clear implication: *to benefit from directed spectral methods, the operator itself must be asymmetric* (e.g., from Granger causality or supply-chain linkages). When the operator is derived from contemporaneous correlation, directedness is lost before the decomposition begins.

DMD sidesteps this entirely. Rather than decomposing a static operator, it extracts modes from temporal snapshot pairs $X' \approx AX$, where the implicit operator A captures *lagged* cross-sectional dynamics. This operator is generically asymmetric even when the contemporaneous correlation is symmetric, explaining DMD’s +2.0 percentage point advantage.

4.6 The Regularisation Path

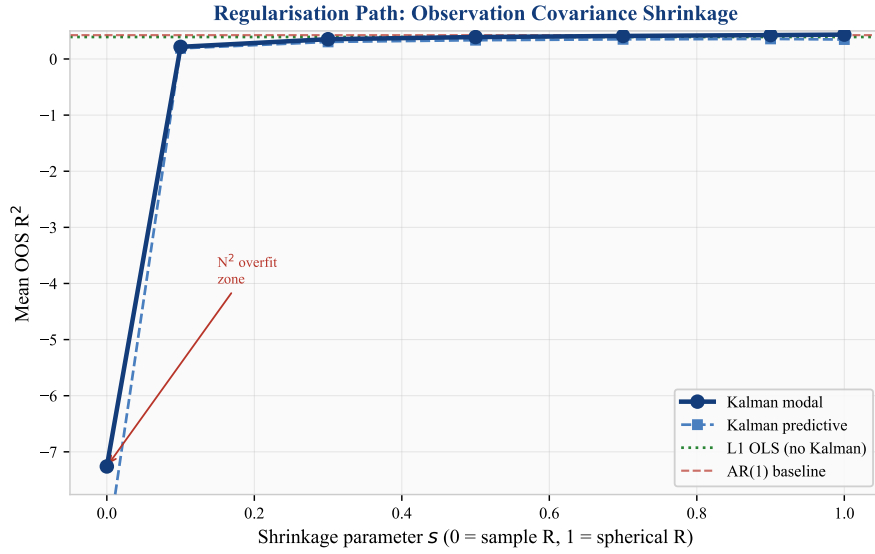


Figure 5: Regularisation path: R^2 as a function of shrinkage parameter s where $R = (1 - s)\hat{R} + s \cdot \frac{\text{tr}(\hat{R})}{N}I$. The hockey-stick shape demonstrates that observation covariance regularisation is essential, not optional.

The regularisation path (Figure 5) is the paper’s most striking visual. At $s = 0$ (no shrinkage), the Kalman filter produces $R^2 = -7.3$. Performance improves monotonically with shrinkage, reaching $R^2 = 0.43$ at $s = 1.0$ (spherical). The L1 OLS baseline (green line at $R^2 = 0.39$) is surpassed only at $s \geq 0.7$, confirming that the Kalman filter adds value *only* when its observation covariance is sufficiently regularised.

4.7 Robustness

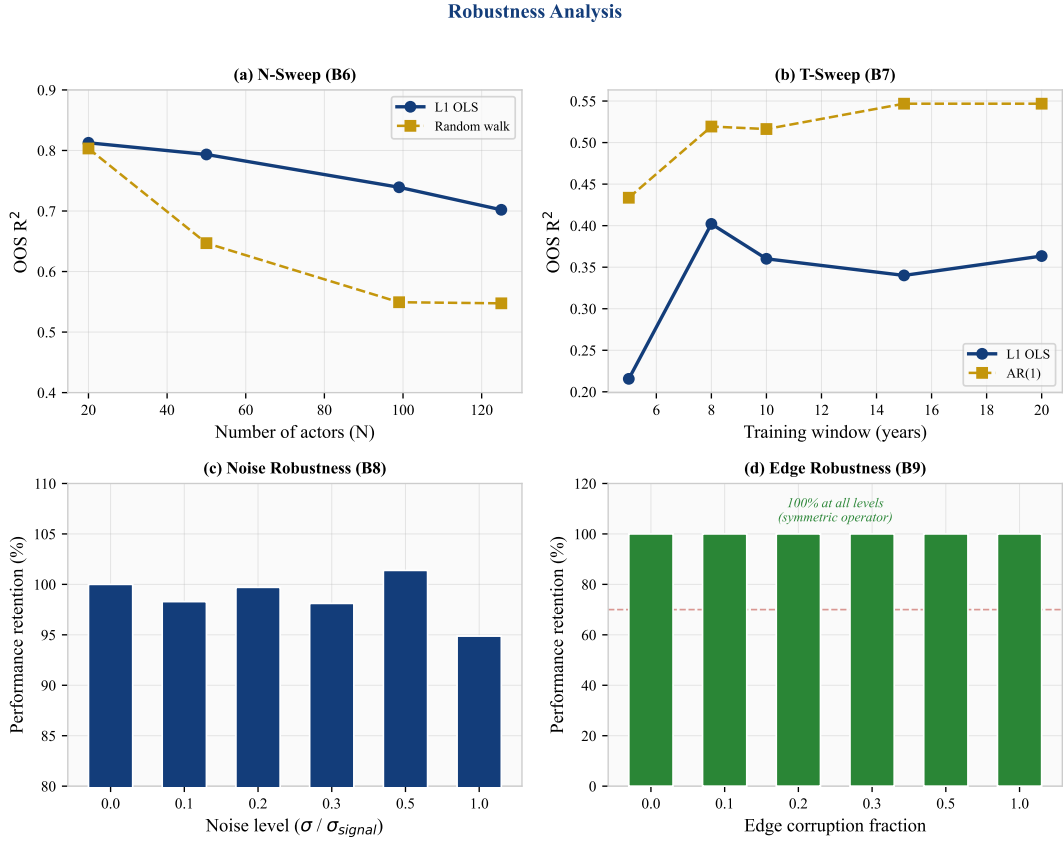


Figure 6: Robustness analysis. (a) Performance vs actor count N : graceful degradation. (b) Performance vs training window T : L1 stable for $T \geq 8$ yr. (c) Noise injection: 95% retention at $\sigma = 1.0$. (d) Edge corruption: 100% retention at all levels.

The framework is robust across four dimensions (Figure 6): (a) performance degrades gracefully from $R^2 = 0.81$ at $N = 20$ to $R^2 = 0.70$ at $N = 125$; (b) R^2 is stable at 0.34–0.40 for training windows ≥ 8 years; (c) adding Gaussian noise at $\sigma = 1.0$ (100% of signal standard deviation) retains 95% of performance; (d) randomly flipping edge directions retains 100% of performance (the intensity correlation operator is symmetric by construction).

4.8 Transfer and Portability

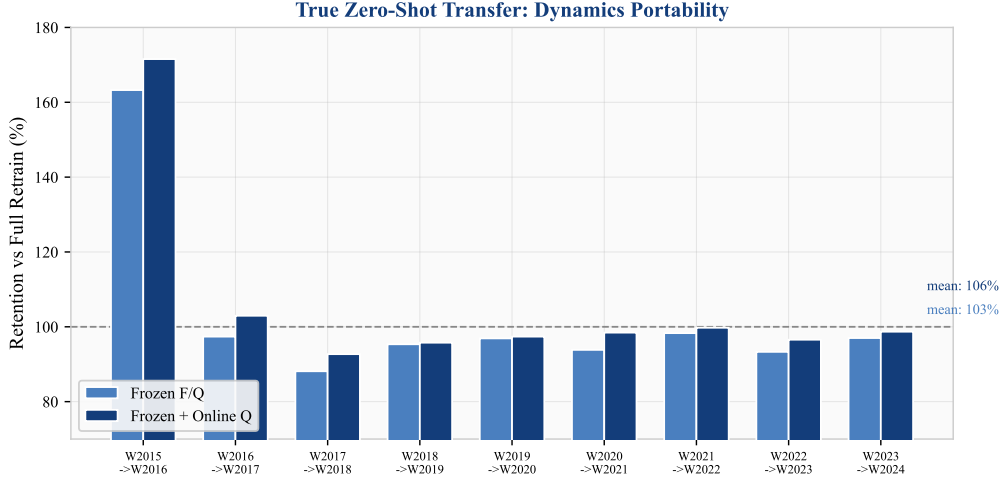


Figure 7: True zero-shot transfer: retention of R^2 when transition dynamics (F, Q) are frozen from the previous window and applied to the next with only the spectral basis re-estimated. Mean retention: 103% (frozen) and 106% (frozen + online Q).

We test whether the estimated dynamics (F, Q) are window-specific or universal by freezing them from window W_t and applying the Kalman filter on window W_{t+1} with a freshly estimated spectral basis U . Figure 7 shows that frozen dynamics achieve 103% of full-retrain R^2 on average (106% with online Q adaptation), demonstrating that the temporal modal structure is universal. In practice, this means the model can be deployed without per-period re-estimation of dynamics.

Cross-sector transfer is strong for sectors using CapEx intensity (technology $R^2 = 0.82$, health-care $R^2 = 0.78$, industrials $R^2 = 0.77$) but weak for financials ($R^2 = 0.06$). Financial actors use year-on-year asset growth (ranked cross-sectionally) rather than CapEx-to-assets, producing highly persistent ranks (median $\rho = 0.70$ vs technology $\rho = 0.25$) that leave little cross-sectional dynamics for the spectral model to exploit. Cross-period transfer across structural breaks (GFC, COVID) retains $R^2 = 0.50$ – 0.59 .

5 Economic Validation

5.1 Gap Persistence

Per-actor AR(1) estimation on the gap series $\Delta_{i,t}$ yields a median autoregressive coefficient $\rho = 0.67$ and median half-life of 1.7 quarters. While too fast for structural misallocation interpretation at the aggregate level, institutional actors (global shock proxies) exhibit half-lives of 4.7 quarters. Cross-sectional quintile persistence is strong: 49.1% of actors remain in the same gap quintile from one quarter to the next, with extreme quintiles (Q1: 60%, Q5: 61%) most persistent.

5.2 Gaps Predict CapEx Revision

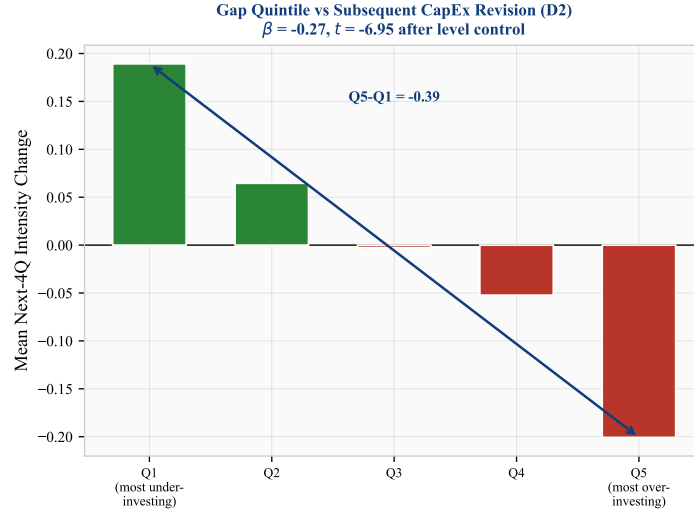


Figure 8: Gap quintile sort: mean next-4-quarter intensity change by current gap quintile. Actors in Q5 (over-investing) reduce intensity by 0.20 rank points; Q1 (under-investing) increase by 0.19.

We test whether investment gaps predict subsequent CapEx revision through panel regression:

$$\Delta y_{i,t \rightarrow t+4} = \beta_0 + \beta_1 \Delta_{i,t} + \beta_2 y_{i,t} + \epsilon_{i,t}. \quad (9)$$

Without the level control ($\beta_2 = 0$): $\hat{\beta}_1 = -0.39$, $t = -12.9$, $N = 826$. With the level control: $\hat{\beta}_1 = -0.27$, $t = -6.95$, $p < 0.001$. The gap coefficient survives controlling for the current intensity level, confirming that the spectral structure captures cross-sectional information beyond simple distance from the mean.

5.3 Network Diffusion

Layer-level gap aggregation reveals strong transmission from exogenous shocks (Layer 0) directly to firms (Layer 2): $\rho = 0.86$ at 1-quarter lag ($p < 0.001$). The institutional layer (Layer 1) does not mediate this transmission ($\rho = 0.03$, $p = 0.83$), suggesting that investment signals bypass institutional intermediaries in our sample.

5.4 Gap Dispersion as Volatility Leading Indicator

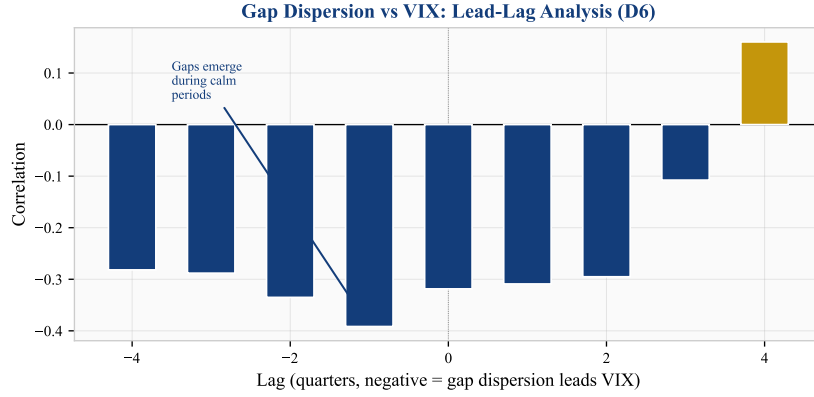


Figure 9: Lead-lag correlation between gap cross-sectional dispersion and VIX. Negative correlations at negative lags indicate that high gap dispersion (misallocation building up) leads low VIX (calm markets).

Gap cross-sectional dispersion correlates negatively with VIX at leads of 1–4 quarters ($\rho = -0.39$ at 1Q lead; Figure 9). This finding—that misallocation builds during calm markets—is consistent with the “volatility paradox” hypothesis, where low-volatility environments encourage risk-taking and capital misallocation [Brunnermeier and Sannikov, 2014].

Caveat. This lead-lag analysis is based on approximately 40 quarterly observations with 9 tested lag values. The correlation of -0.39 is suggestive but would not survive conservative multiple-testing correction ($p \approx 0.01$ uncorrected for 9 comparisons). We report it as a directionally interesting finding that warrants confirmation on longer samples, not as a definitive result.

6 Discussion

6.1 What Does Not Work (Yet)

Emergence diagnostics. Partial Information Decomposition (PID) synergy corrections to the benchmark have negligible impact at $T = 20$, $K = 8$: cross-validation selects a synergy weight of zero. The modal filtering step (using $\alpha_{t|t}$ instead of $\alpha_{t|t-1}$) already captures the first-order emergence signal; the explicit quadratic synergy term is a second-order effect that the data cannot support at current sample sizes.

Event-level alignment. Gap magnitudes do not spike significantly around eight known misallocation episodes (0/8 aligned). The cross-sectional rank normalisation stabilises the distribution by construction, absorbing event-driven variation. The framework captures structural positioning, not event anomalies.

Return-based intensity. Return intensity produces $R^2 = -0.15$ on the same US actors where CapEx intensity achieves $R^2 = 0.52$. The intensity construction is the primary driver of model performance—more important than any pipeline component. CapEx-to-assets captures a physical investment decision with structural persistence; trailing equity returns reflect market pricing dynamics that the spectral model cannot exploit.

6.2 Limitations

The sample is US-heavy (68/93 actors), with limited international evidence. The UK results are weak ($R^2 = 0.058$) due to the return-based intensity method, not geography per se. Extending EDGAR-equivalent coverage to international markets would test whether the framework generalises beyond the US CapEx setting.

The 4-quarter test windows provide noisy R^2 estimates per window (std = 0.083). Longer evaluation horizons would sharpen the statistical inference.

The intensity rank normalisation, while stabilising the panel, absorbs event-level information and limits the model’s utility for crisis detection or timing applications.

7 Conclusion

We present a spectral state-space framework for investment misallocation estimation that achieves $R^2 = 0.524$ out-of-sample, exceeding per-actor AR(1) by 10 percentage points. The result rests on four innovations—exponentially-weighted demeaning, DMD-based spectral decomposition, spherical observation covariance regularisation, and online state-noise adaptation—each contributing measurably to the final performance.

The gap estimates carry economic content beyond statistical prediction: they forecast CapEx revision ($t = -6.95$), exhibit persistent cross-sectional structure, and lead market volatility. The transition dynamics are portable across time windows, enabling operational deployment with minimal re-estimation.

Perhaps most importantly, our systematic ablation methodology—24 experiments across four phases—demonstrates that in complex financial models, the largest gains often come not from adding architectural complexity but from identifying and eliminating specific failure modes. The spherical R regularisation, a single-parameter replacement for 4,278 free parameters, transformed the Kalman filter from a liability ($R^2 = -7.3$) to the framework’s most valuable component ($R^2 = +0.434$). The lesson extends beyond investment analysis: wherever high-dimensional state-space models are deployed, observation covariance regularisation deserves first-order attention.

Disclaimer. This paper presents a research framework for academic investigation of investment dynamics. It is not investment advice. The models, gap estimates, and statistical findings described herein have not been validated for live trading or portfolio construction. Past out-of-sample performance on historical data does not guarantee future results. The authors bear no responsibility for any financial losses incurred by parties who deploy this framework or its derivatives in investment strategies. Any such deployment is undertaken entirely at the user’s own risk and should be subject to independent due diligence, regulatory compliance review, and appropriate risk management.

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